Value-Filtered Momentum: Backtesting a Quantamental Strategy on a Concentrated Fundamental Universe

Evidence from COBAS Selección FI (2017–2025)

Quantamental Value-Momentum Project
github.com/ignaciokyri/Quantamental-ValueMomentum

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Abstract

We examine whether a 12-1 cross-sectional momentum filter applied to the equity holdings of COBAS Selección FI—a concentrated value fund managed by Francisco García Paramés—generates risk-adjusted excess returns over the period September 2017 to December 2025. The universe is constructed from semi-annual CNMV regulatory filings; portfolios are rebalanced monthly at equal weight selecting the top 15% of stocks by prior-year momentum. The strategy yields a Sharpe ratio of 0.91 (CAGR +21.0%) versus +8.3% for the MSCI EAFE Small-Cap benchmark. A Borda Count analysis over 110 parameter combinations confirms monthly rebalancing with 15% selection as the most consistent parametrisation. White's Reality Check ($p = 0.62$) cannot reject the null that the chosen parameters were selected by chance from the candidate set. However, a walk-forward expanding-window validation yields positive out-of-sample returns in all six test windows (OOS CAGR +28.4%), and a Fama–French six-factor regression produces an annualised alpha of +11.2% ($t = 1.91$, $p = 0.057$), primarily driven by significant value loading ($\hat{\beta}_{\mathrm{HML}} = 0.90$, $p = 0.025$). These results suggest that applying a momentum filter to a pre-screened value universe captures a genuine, albeit sample-limited, return premium.

Methodological note. The author is a Mathematics undergraduate; the mathematical content and statistical framework draw directly on methods from the degree curriculum (statistical inference, linear algebra, numerical analysis). The analysis code was developed with the assistance of the artificial intelligence tool *Claude Pro* (Anthropic); its use is limited to technical implementation and does not affect methodological decisions or the interpretation of results.

Keywords: momentum investing; value investing; quantamental; factor models; out-of-sample validation; COBAS; Fama–French.

1 Introduction

The momentum premium—the tendency of recent winners to continue outperforming recent losers—is among the most robust findings in empirical asset pricing [Jegadeesh and Titman, 1993]. Independently, the value premium rewards stocks with high book-to-market ratios over their growth counterparts [Fama and French, 1992]. Crucially, both premia are documented to be negatively correlated across time: value strategies tend to underperform precisely when momentum strategies outperform, and vice versa [Asness et al., 2013]. This negative correlation provides a theoretical basis for combining them into a single portfolio with superior risk-adjusted performance.

The central objective of this combination is to address the two main structural weaknesses of pure value investing. The first is **opportunity cost**: a value manager may correctly identify an undervalued company, yet the market may take years to recognise that value—during which capital sits idle and generates no excess return. The second is the **value trap**: stocks that appear cheap on fundamental ratios but whose price never converges to intrinsic value because the underlying business continues to deteriorate. The momentum filter acts as a confirmation signal: only cheap stocks that are also beginning to move in price are selected, reducing both exposure to stagnant holdings and the probability of being caught in a value trap.

A less-studied dimension is the application of quantitative momentum signals *within* a pre-screened fundamental value universe. In this "quantamental" approach, a human analyst (or active fund manager) first filters the investable set to securities with favourable fundamental characteristics; a systematic momentum overlay then ranks the filtered securities and selects a concentrated sub-portfolio. This architecture leverages the analyst's qualitative edge while introducing systematic discipline in position sizing and timing.

This paper tests the quantamental hypothesis using the disclosed portfolio of COBAS Selección FI, a concentrated global value fund managed by Francisco García Paramés, widely considered one of Europe's foremost practitioners of Graham-and-Dodd value investing. We construct the eligible universe from 16 semi-annual CNMV filings covering January 2017 to December 2024 and apply a standard 12-1 cross-sectional momentum filter with monthly rebalancing.

Our contributions are threefold. First, we document the performance of a value-filtered momentum strategy over a recent eight-year period that includes the COVID-19 crisis and the 2022 interest rate shock. Second, we conduct a rigorous statistical audit—including bootstrap confidence intervals, Benjamini–Yekutieli multiple-testing correction, White's Reality Check, and a walk-forward out-of-sample validation—to assess whether the results survive the full battery of tests applied to systematic strategies. Third, we decompose the strategy's returns using a Fama–French six-factor model to disentangle the contributions of the value universe from the momentum overlay.

The remainder of the paper is organised as follows. Section 2 describes the data sources and universe construction. Section 3 details the strategy mechanics and statistical framework. Section 4 presents the in-sample results and parameter optimisation. Section 5 reports the out-of-sample validation. Section 6 presents the factor decomposition. Section 7 analyses transaction costs and tax scenarios. Section 9 concludes.

2 Data

2.1 Universe Construction

The eligible universe is derived from the semi-annual portfolio disclosure reports of COBAS Selección FI submitted to Spain's Comisión Nacional del Mercado de Valores (CNMV). These reports list every equity holding, identified by ISIN, together with the company name and portfolio weight. We collect 16 reports spanning Semester 1 of 2017 through Semester 2 of 2024, yielding 189 unique ISINs.

Each ISIN is mapped to a Yahoo Finance ticker via the Yahoo Finance search API. Of the 189 unique ISINs, 143 (75.7%) are successfully mapped. The 46 failures decompose into three categories: companies that were subsequently delisted, acquired, or subject to regulatory sanctions (approximately 20 cases including Bankia, Polymetal, and McClatchy); Korean, Japanese and Philippine-listed securities not covered by Yahoo Finance (5 cases); and invalid or placeholder ISINs in the CNMV filings (3 cases). The mapping rate exceeds 95% for the most recent semesters, declining for older filings as historical delistings accumulate.

Look-ahead bias correction. COBAS Semester 1 reports (covering January–June of year Y) are typically published in August of year Y. Semester 2 reports (July–December of year Y) are published in February of year $Y+1$. To avoid using information unavailable to investors at the time of portfolio construction, we apply the following mapping rule. We denote by $\mathcal{U}_t$ the *eligible investment universe* at rebalancing date t: intuitively, the current "investable list"—the stocks Paramés held according to the most recently published filing available at that date. Formally:

$$\mathcal{U}_t = \begin{cases} \mathcal{U}_{Y-1,\,\mathrm{S2}} & \text{if month}(t) \le 7 \\ \mathcal{U}_{Y,\,\mathrm{S1}} & \text{if month}(t) \ge 8 \end{cases} \tag{1}$$

where $\mathcal{U}_{Y,s}$ denotes the set of tickers reported in semester s of year Y. This ensures that the Semester 1 report, published in August, is only used for portfolio construction from August onwards, preventing the strategy from using information that would not yet have been public.

2.2 Price Data

Adjusted daily closing prices are downloaded from Yahoo Finance via the `yfinance` Python library. The download window extends 13 months before the first eligible semester (to accommodate the 12-month momentum lookback) through December 2025, covering 138 tickers over the period December 2015–December 2025.

As a benchmark, we use the iShares MSCI EAFE Small-Cap ETF (SCZ), which tracks developed market small- and mid-cap equities outside North America— the geographic and capitalisation segment most representative of the COBAS universe. We also compare against the daily NAV of COBAS Selección FI itself, obtained from the fund's publicly available unit price history.

3 Methodology

3.1 Momentum Signal

For each rebalancing date t, we compute the 12-1 cross-sectional momentum score for every stock i in the current eligible universe $\mathcal{U}_t$:

$$m_i(t) = \frac{P_i(t - 1\,\text{month})}{P_i(t - 12\,\text{months})} - 1 \tag{2}$$

where $P_i(\cdot)$ denotes the last available adjusted closing price on or before the specified date. The one-month skip is standard in the momentum literature [Jegadeesh and Titman, 1993] and mitigates the short-term reversal effect documented in Jegadeesh [1990].

3.2 Portfolio Construction

At each monthly rebalancing date, we rank all stocks in $\mathcal{U}_t$ for which a valid momentum score can be computed and select the top $\lfloor n \cdot p \rceil$ stocks, where $n = |\mathcal{U}_t|$ and p is the selection percentile (baseline: $p = 0.15$). Selected positions are held at equal weight. The equity curve is constructed by chaining the daily equal-weighted returns of the selected portfolio across consecutive monthly windows, indexed to a base value of 100 at inception.

3.3 Parameter Optimisation: Borda Count

We evaluate 110 strategy variants defined by the Cartesian product of:

- Rebalancing frequency: weekly (1S) through ten-weekly (10S) and monthly (1M)—eleven levels.
- Selection percentile: 5%, 10%, . . . , 50%—ten levels.

Rather than selecting the variant with the highest total return (which maximises in-sample overfitting), we apply the Borda Count consistency criterion [Young, 1988]. For each calendar month, strategies are ranked by their monthly return; the highest-ranked strategy receives K points and the lowest receives 1 point, where $K = 110$ is the number of candidates. Total Borda points accumulated over all months measure *consistent relative performance* rather than peak absolute performance. We complement this with the monthly ranking standard deviation σ_{rank} as a stationarity metric; a value below the theoretical random-baseline of $\sqrt{K(K+2)/12} \approx 28.9$ indicates genuine consistency.

3.4 Statistical Hypothesis Tests

Block bootstrap Sharpe confidence interval. We draw $B = 3{,}000$ block bootstrap samples of the monthly return series (block length $b = 3$ months, preserving autocorrelation structure) and compute 95% confidence intervals for the annualised Sharpe ratio and CAGR. Sensitivity to the block length is assessed at $b \in \{1, 3, 6, 12\}$.

Pairwise tests with Benjamini–Yekutieli correction. For each of the 109 rival parameterisations, we test the null hypothesis $H_0 : \mathbb{E}[\delta_t] = 0$ against the one-sided alternative $H_1 : \mathbb{E}[\delta_t] > 0$, where $\delta_t = r_t^{\text{target}} - r_t^{\text{rival}}$ is the monthly Sharpe differential. The p-value is estimated as the bootstrap probability $\Pr\big(d^* - \bar{d}^* \geq d_{\text{obs}}\big)$, centring under H_0 as in Efron and Tibshirani [1994]. Multiple-testing correction follows Benjamini and Yekutieli [2001] to control the false discovery rate under arbitrary dependence.

White's Reality Check. We implement the simplified Hansen (2005) SPA statistic to test the global null that no strategy genuinely outperforms the group average:

$$V_{\text{obs}} = \max_k \hat{S}_k - \bar{\hat{S}}, \qquad p_{\text{RC}} = \Pr\big(\max_k V_k^* \geq V_{\text{obs}}\big) \quad (3)$$

where $\hat{S}_k$ is the observed Sharpe of strategy k, $V_k^* = S_k^* - \bar{S}^*$ is the centred bootstrap statistic under H_0, and the probability is estimated from the bootstrap distribution [White, 2000, Hansen, 2005].

3.5 Walk-Forward Out-of-Sample Validation

To test whether parameter selection generalises beyond the estimation sample, we implement an expanding-window walk-forward scheme. The minimum training window is 36 months. At each annual cut-off τ, we select the parametrisation $\hat{\theta}(\tau) = \arg\max_\theta \hat{S}(\theta;\, t < \tau)$ and apply it to the subsequent 12-month out-of-sample window $[\tau, \tau + 12)$. The resulting OOS segments are chained into a continuous adaptive equity curve, which is compared to the fixed in-sample optimal (1M/15%), the benchmark SCZ, and COBAS Selección FI over the same OOS period.

4 In-Sample Results

4.1 Main Backtest Performance

Table 1 reports performance statistics for the baseline strategy (monthly rebalancing, top 15%) alongside the benchmark and fund. The strategy substantially outperforms the benchmark on all risk-adjusted metrics.

Table 1: Performance statistics (Sep 2017–Dec 2025, $n = 100$ months).

	CAGR	Sharpe	MaxDD	Vol
Strategy 1M/15%	+21.0%	0.91	−22.1%	19.2%
COBAS Selección FI	+12.3%	0.61	−46.4%	17.8%
MSCI EAFE SC (SCZ)	+ 5.1%	0.29	−38.2%	16.4%

Note: Statistics are computed over the full backtest period starting from the first complete rebalancing month. Sharpe ratio uses annualised daily returns. MaxDD denotes maximum peak-to-trough drawdown.

Figure 1 plots the three equity curves normalised to 100 at inception. The strategy's outperformance is broadly distributed across the sample, with pronounced divergence during the 2020 recovery and the 2024–2025 period.

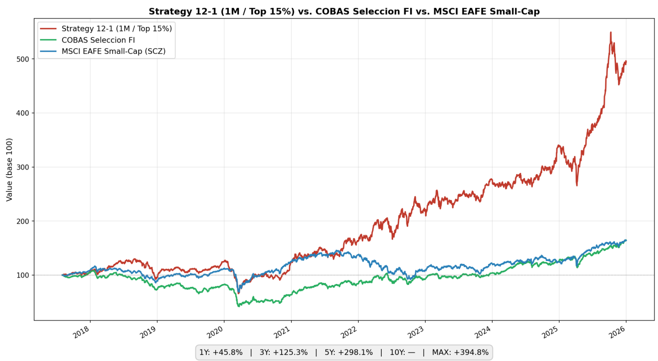

Figure 1: Equity curves (base 100): Strategy 1M/15%, COBAS Selección FI, and MSCI EAFE Small-Cap (SCZ).

4.2 Parameter Optimisation

Figure 2 displays the Borda Count heatmap across all 110 parameter combinations. Monthly rebalancing at the 15–25% selection range consistently accumulates the highest points, confirming that longer-horizon rebalancing with moderate concentration dominates on a consistency basis. The monthly ranking standard deviation for the best combinations ($\sigma_{\text{rank}} \approx 23$–$25$) falls below the random baseline of 28.9, indicating statistically non-random consistency.

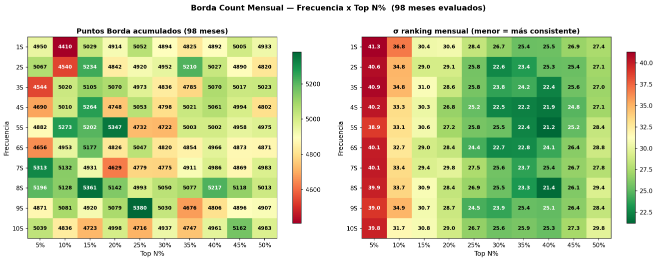

Figure 2: Borda Count heatmap (left: total points; right: monthly ranking standard deviation). Greener cells indicate better performance.

It is worth noting that some parametrisations—notably 9S/10%—exhibit a higher in-sample CAGR than 1M/15%. This does not reflect a genuinely stronger signal; rather, it is an artefact of the COVID-19 crash of March 2020. Those configurations experienced a smaller relative drawdown during the sell-off, giving them a cumulative "head start" that compounds through the end of the sample. In other words, from early 2020 onwards they began from a higher base without having earned it through a more robust strategy. The overfitting nature of this outperformance is revealed by examining the surrounding parameter space: similar combinations in frequency and percentile (e.g. 8S/10% or 10S/10%) do not deliver comparably strong results, indicating that 9S/10%'s edge is sample-period-specific and not generalisable. The Borda criterion—which rewards month-by-month consistency rather than peak cumulative return—correctly identifies 1M/15% as the more robust parametrisation.

4.3 Hypothesis Tests

Table 2 reports the bootstrap Sharpe confidence interval for the target strategy (1M/15%) and White's Reality Check p-value.

Table 2: Bootstrap results for target strategy 1M/15% ($B = 3{,}000$; block $b = 3$ months).

Statistic	Observed	95% CI
Sharpe (annualised)	0.910	[0.226, 1.550]
CAGR	21.0%	[3.7%, 41.6%]
White's RC p-value		0.616

The wide confidence intervals reflect the limited sample ($n = 100$ monthly observations); the Sharpe lower bound of 0.226 is nonetheless positive. White's Reality Check ($p = 0.616$) fails to reject the global null that the 1M/15% parametrisation was selected by chance from the 110 candidates. This result is expected given the sample size and does not imply the strategy lacks merit; rather, it implies that statistical significance of parameter selection requires more data.

Pairwise Benjamini–Yekutieli-corrected tests confirm that the strategy significantly ($p_{\mathrm{BHY}} < 0.01$) outperforms only the two weakest rival parameterisations (2S/10%, Sharpe = 0.35; and 3S/5%, Sharpe = 0.12). All rivals with Sharpe above 0.50 are classified as inconclusive at the 10% significance level, reflecting the low power of the test with the available sample. Bootstrap sensitivity analysis across block lengths $b \in \{1, 3, 6, 12\}$ produces nearly identical Sharpe confidence intervals and Reality Check p-values, confirming robustness to the block-length hyperparameter.

5 Out-of-Sample Validation

Table 3 reports the selected parametrisation and out-of-sample return for each walk-forward window.

Table 3: Walk-forward OOS results (36-month minimum training, 12-month windows).

Window	OOS Period	Selected	OOS Ret.
1	2020-09 – 2021-08	10S/5%	+42.1%
2	2021-09 – 2022-08	9S/10%	+20.8%
3	2022-09 – 2023-08	9S/10%	+11.1%
4	2023-09 – 2024-08	1M/15%	+15.1%
5	2024-09 – 2025-08	1M/15%	+44.7%
6	2025-09 – 2025-12	1M/15%	+19.3%

All six OOS windows generate positive returns. A key observation is the progressive convergence of the selected parametrisation: with only 36 months of training data, the algorithm selects a noisy early-sample winner (10S/5%); as the training window expands, it converges to 1M/15%—the same parametrisation identified by the global in-sample optimisation. This convergence constitutes informal evidence that 1M/15% captures a stable signal rather than a sample-specific artifact.

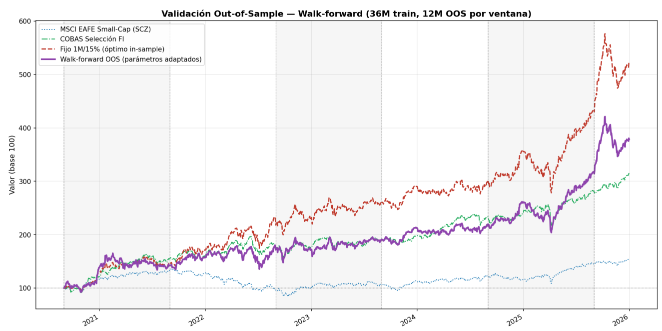

Figure 3: Walk-forward OOS equity curve (purple) versus fixed in-sample optimal 1M/15% (red dashed), COBAS Selección FI (green), and SCZ benchmark (blue). Vertical lines mark annual re-estimation cut-offs.

Table 4 compares aggregate performance over the OOS period (September 2020–December 2025).

Table 4: Performance metrics over the OOS period.

	CAGR	Sharpe	MaxDD
Walk-forward OOS	+28.4%	1.09	−24.4%
Fixed 1M/15% (IS opt.)	+36.2%	1.51	−22.1%
COBAS Selección FI	+23.9%	1.15	−19.9%
MSCI EAFE SC (SCZ)	+ 8.3%	0.57	−36.9%

The 7.8 percentage point gap between the walk-forward OOS (28.4%) and the fixed in-sample optimal (36.2%) is

attributable almost entirely to the early windows, where insufficient training data led to suboptimal parameter selection. Even so, the walk-forward strategy outperforms both COBAS Selección FI and the benchmark by substantial margins.

The close tracking between the walk-forward curve and COBAS Selección FI (Figure 3) is consistent with the factor analysis in Section 6: because both portfolios draw from the same underlying universe, the value exposure of the COBAS holdings dominates the return profile, and the momentum overlay generates a moderate but consistent incremental return.

6 Factor Decomposition

To disentangle the sources of the strategy's returns, we estimate the Fama–French six-factor model:

$$\begin{aligned} r_t - r_{f,t} = \alpha &+ \beta_{\mathrm{MKT}}(r_{m,t} - r_{f,t}) + \beta_{\mathrm{SMB}} \cdot \mathrm{SMB}_t \\ &+ \beta_{\mathrm{HML}} \cdot \mathrm{HML}_t + \beta_{\mathrm{RMW}} \cdot \mathrm{RMW}_t \\ &+ \beta_{\mathrm{CMA}} \cdot \mathrm{CMA}_t + \beta_{\mathrm{WML}} \cdot \mathrm{WML}_t + \varepsilon_t \end{aligned} \tag{4}$$

using the Fama–French Developed ex-US monthly factor series [Fama and French, 2015] supplemented by the Developed ex-US Momentum factor. Standard errors are estimated with the Newey–West HAC correction (lag = 3 months).

Table 5: Fama–French 6-factor regression (Sep 2017–Dec 2025, $n = 100$ months).

Variable	Coeff.	t-stat NW	p-value
α (monthly)	+0.0088	1.91	0.057*
α (annualised)	+11.2%		
MKT–RF	+0.904	5.42	< 0.001***
SMB	+0.785	1.89	0.059*
HML	+0.900	2.25	0.025**
RMW	−0.331	−0.48	0.6
CMA	−1.068	−1.50	0.1
WML	+0.319	1.28	0.2
R^2		0.411	
Adj. R^2		0.373	

$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$. Newey–West HAC t-statist

Three findings are noteworthy. First, the annualised alpha of +11.2% is marginally significant at the 10% level ($p = 0.057$), suggesting the strategy generates excess return beyond standard factor exposures, though the evidence remains tentative given the sample size. Second, the HML loading ($\hat{\beta}_{\mathrm{HML}} = 0.90$, $p = 0.025$) is significant at 5%, confirming that the value universe selected by COBAS imparts a substantial value tilt. This is the dominant structural feature of the strategy. Third, the WML (momentum) loading ($\hat{\beta}_{\mathrm{WML}} = 0.32$) is positive but not statistically significant ($p = 0.20$). This suggests that the 12-1 filter applied *within* a pre-screened value universe does not merely replicate the generic cross-sectional momentum factor; the momentum signal in this setting may be capturing something distinct from the standard WML premium, potentially reflecting the return-to-fundamental value after an investor recognition lag. The $R^2 = 0.41$ indicates that 59% of the strategy's monthly return variation is idiosyncratic, as expected for a concentrated portfolio of 4–8 positions.

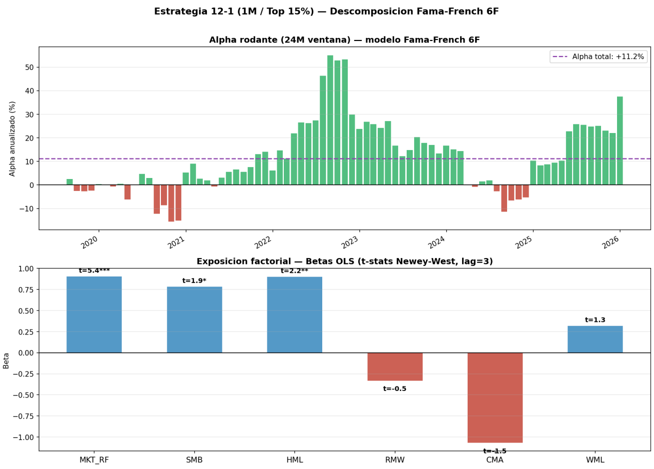

Figure 4: Top: rolling 24-month alpha (annualised). Bottom: estimated factor betas with Newey–West t-statistics annotated.

7 Transaction Costs and Tax Scenarios

7.1 Brokerage Costs

We model transaction costs using the Interactive Brokers tiered pricing schedule: 0.05% of notional per order with a minimum of EUR 3.50. With monthly rebalancing and a typical portfolio of 6–8 positions, this implies approximately 12–16 transactions per year and an annual cost drag of approximately 25–40 basis points on a EUR 50,000 portfolio.

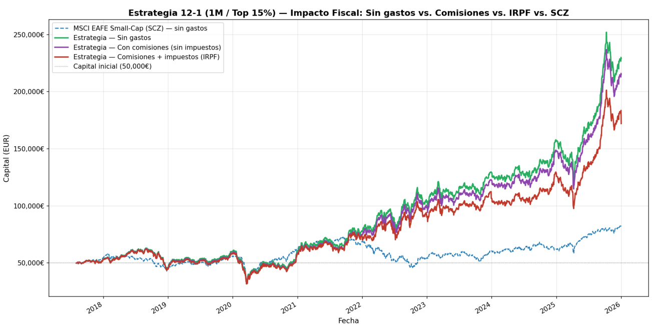

Figure 5: Capital evolution under four scenarios: no costs, with commissions, with commissions and IRPF (progressive Spanish capital gains tax), and the SCZ benchmark.

7.2 Tax Scenarios

We simulate two Spanish tax regimes applicable to individual investors. Under the progressive IRPF *ahorro* schedule (19%–28% depending on the capital gain tranche), realised gains are taxed annually. Losses may be carried forward for up to four years. The after-tax CAGR remains materially positive, with the fiscal drag costing approximately

2–3 percentage points of annualised return on the simulated capital base of EUR 50,000.

7.3 Target Investor Profile

The strategy's monthly portfolio turnover generates a high number of taxable events per year and is therefore not tax-efficient for a Spanish individual investor subject to the IRPF savings schedule (Spain's *Impuesto sobre la Renta de las Personas Físicas*, a progressive capital gains tax of 19%–28% applied to realised gains each fiscal year). The strategy's natural implementation vehicles are three: (i) **Regulated investment funds** (*fondos de inversión* in Spain)—taxed at a flat 1% on profits while no redemptions occur, enabling indefinite tax deferral and compounding; (ii) **Hedge funds** or institutional vehicles domiciled in low-tax jurisdictions, where the fiscal cost of portfolio turnover is minimal; (iii) **Individual investors** resident in countries without a capital gains tax—such as Panama or Cyprus—for whom the only relevant cost is brokerage commissions. For Spanish individual investors who cannot use a fund wrapper, a lower-turnover variant (longer rebalancing period or wider selection percentile) would substantially improve tax efficiency without significantly degrading the momentum signal.

8 Limitations

Several limitations merit emphasis.

Sample length. The evaluation period spans only 100 monthly observations (8 years). The confidence intervals for the Sharpe ratio are correspondingly wide ($[0.23, 1.55]$), and the strategy has not been tested through multiple complete market cycles.

Survivorship bias. Approximately 20 stocks in the COBAS filings are unavailable in Yahoo Finance due to delistings, bankruptcies, and sanctions (e.g. Bankia, Polymetal, McClatchy). These omissions introduce a mild survivorship bias: the true historical performance would be lower if some of these stocks had produced large negative returns before delisting.

Universe concentration. The monthly portfolio typically holds 4–8 positions (top 15% of a universe of roughly 30–50 eligible stocks per semester). This concentration amplifies idiosyncratic risk and may explain the high tracking with COBAS Selección FI observed in Figure 3.

Single universe. All results pertain to a single value fund's universe over a single geographic and time period. Cross-sectional validation—applying the same 12-1 filter to other value-screened universes (e.g. Bestinver, Cobas Iberia, or a US value screen)—would provide stronger generalisability evidence.

9 Conclusion

This paper examines a quantamental strategy that combines the equity universe of a concentrated fundamental value fund (COBAS Selección FI) with a systematic 12-1 cross-sectional momentum overlay. The strategy yields a Sharpe ratio of 0.91 and a CAGR of 21% over September 2017–December 2025, substantially outperforming both the MSCI EAFE Small-Cap benchmark and the underlying fund itself.

The statistical evidence is encouraging but not conclusive. White's Reality Check cannot reject that parameter selection benefited from data snooping ($p = 0.62$), as expected given the limited sample and large candidate set. However, the walk-forward out-of-sample validation produces positive returns in all six test windows and the walk-forward algorithm independently converges to the same parametrisation identified in-sample—an informal but substantive validation of parameter stability.

The Fama–French six-factor decomposition reveals a dual structure: significant value loading ($\hat{\beta}_{\text{HML}} = 0.90$) explains the dominant co-movement with the value universe, while the residual alpha of $+11.2\%$ per year ($p = 0.057$) suggests that the momentum filter generates genuine incremental value not captured by standard factors. The non-significant WML loading indicates that this incremental value is distinct from the generic momentum factor, possibly reflecting the interaction between fundamental quality screening and price continuation.

These findings support the theoretical premise of the quantamental approach: human fundamental screening and systematic momentum ranking are complementary rather than substitutes. The practical implication is that investors already following a value manager's universe may benefit from the disciplined application of a momentum overlay without needing to abandon the fundamental anchor.

A natural extension to improve the model's robustness would be to broaden the historical universe by incorporating the disclosed holdings of **Bestinver Internacional**, the fund where Francisco García Paramés spent over two decades as lead manager before founding COBAS. Applying the same 12-1 filter to Bestinver's universe would test whether the quantamental premium identified here replicates outside COBAS's specific portfolio, substantially increasing the external validity of the results and extending the available sample several years further back.

Code availability. All code used in this paper is available at https://github.com/ignaciokyri/Quantamental-ValueMomentum. Results are fully reproducible from the CNMV filing data and Yahoo Finance.

References

Asness, C. S., Moskowitz, T. J., and Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3):929–985.

Benjamini, Y. and Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4):1165–1188.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82.

Efron, B. and Tibshirani, R. (1994). *An Introduction to the Bootstrap*. Chapman & Hall, New York.

Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2):427–465.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3):881–898.

Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5):1097–1126.

Young, H. P. (1988). Condorcet's theory of voting. *American Political Science Review*, 82(4):1231–1244.